Test Fri

Ex 1. a) $z = 12$

$l = 12$

$v = (2 \cdot z + 3 \cdot l) \bmod 11 = (24 + 36) \bmod 11 = 60 \bmod 11 = 5$

b) $x \equiv (v+1) \bmod 7$

$x \equiv (v+2) \bmod 5$

$x \equiv (v+3) \bmod 13$

$x \equiv b \bmod m$

$\Rightarrow x \equiv 6 \bmod 7$

$x \equiv 7 \bmod 5$

$x \equiv 8 \bmod 13$

$m = 7 \cdot 5 \cdot 13 = 455$

$c_1 = m / m_1 = 455 / 7 = 65$

$c_2 = m / m_2 = 455 / 5 = 91$

$c_3 = m / m_3 = 455 / 13 = 35$

$65x \equiv 6 \bmod 7$ $\quad x_1 = 3$

$91x = 7 \bmod 5$ $\quad x_2 = 2$

$35x = 8 \bmod 13$ $\quad x_3 = 11$

$x = (c_1 \cdot x_1 + \ldots + c_k x_k) \bmod m = (65 \cdot 3 + 91 \cdot 2 + 35 \cdot 11) \bmod 455$

$= (195 + 182 + 385) \bmod 455 = 762 \bmod 455 = 307.$

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Ex. 2. Inverse modulare

a) 5^{-1} mod 19

5^{-1} mod 19

$a = 5 \quad m = 19$

$(5, 19) = (19, 5)$ $\quad V_a = (1,0) \quad V_b = (0,1)$

$19 = 3 \cdot 5 + 4$ $\quad V_{r_1} = V_a - q_1 V_b = (1,0) - 3(0,1) = (1,-3)$

$5 = 1 \cdot 4 + 1$ $\quad V_{r_2} = V_b - q_2 V_{r_1} = (0,1) - 1(1,-3) = (-1, 4)$

$4 = 4 \cdot 1$ $\quad \alpha = 4 \quad \beta = -1$ (le-am inversat la input)

$(a, m) = 5 \cdot 4 + (-1) \cdot 19 = 20 - 19 = 1. \rightarrow$

$\rightarrow 5^{-1} = 4$ mod 19

b) $(-5 \text{ mod } 11)^{-1}$ mod 19

~~-5 mod 11 , 11-5 = 6.~~

-5 mod 11 $= (11-5)$ mod 11 $= 6$ mod 11 $= 6. \rightarrow 6^{-1}$ mod 19

$a = 6 \quad m = 19$

$(6, 19) = (19, 6)$

$19 = 3 \cdot 6 + 1.$ $\quad V_{r_1} = V_a - q_1 V_b = (1,0) - 3(0,1) = (1,-3)$

$\alpha = -3 \quad \beta = 1.$

$6 = 6 \cdot 1$

$(a, m) = 6 \cdot (-3) + 19 \cdot 1 = -18 + 19 = 1.$

$\rightarrow 6^{-1}$ mod 19 $= -3$ mod 19 $= 16$ mod 19

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